Technology Stack (Architecture & Stack)

Date	28 October 2026
Team ID	EN25CS3010362
Project Name	Formula 1 Race Intelligence Dashboard
Maximum Marks	As per supplied template

Defined Technology Stack

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language used for web application and analytics expressions.	Python; Tableau calculated fields; HTML/CSS	Python supports Flask. Tableau calculated fields may support workbook measures. SQL use is not confirmed.
2	Database / Data Warehouse	Structured data source used by the workbook.	Tableau-connected dataset (exact format not confirmed)	The evidence confirms a Tableau workbook but does not confirm Snowflake, MySQL, or another warehouse. Record the actual source format before submission.
3	Data Ingestion	Load data into the analysis environment.	Tableau data connection	Tableau connects to the dataset used by the workbook. A separate ingestion pipeline or Snowflake stage is not evidenced.
4	Data Transformation and Business Logic	Data preparation, calculated measures, aggregation, and filtering.	Tableau workbook calculations and filters; Python for website	The dashboard uses Tableau charts and filters. Exact transformations should be documented from the workbook.
5	Data Visualization / Analytics	Interactive charts, KPI cards, maps, and story views.	Tableau Public	Publishes the interactive dashboard and Story.
6	Front-End / User Interface	Website and dashboard entry points.	Flask; HTML/CSS; Tableau Public	Flask website hosted on Render links users to the published Tableau views.

Project URLs

Website: https://f1-race-intelligence-mhgm.onrender.com/

GitHub: https://github.com/gourisanjaysharma-source/F1-Race-Intelligence

Tableau Dashboard:
https://public.tableau.com/app/profile/gouri.sharma8417/viz/f1tableauproject/F1RACEINTELLIGENCEDASHBOARD

Tableau Story: https://public.tableau.com/views/f1tableauproject/Formula1RaceIntelligence